Lecture 3: Practice Problems — Answers and Hints

Math for Machine Learning

October 23, 2026

Multivariable Functions

1.1. (a) $\mathbb{R}^2$; the parallel lines $2x - y = c - 1$, with normal vector $(2, -1)$. (b) $\mathbb{R}^2$; for $c > 0$ the ellipses $\frac{x^2}{c/4} + \frac{y^2}{c} = 1$ with half-axes $\frac{\sqrt{c}}{2}$ and $\sqrt{c}$, the origin for $c = 0$, and the empty set for $c < 0$. (c) $\mathbb{R}^2$; the parabolas $y = x^2 + c$, which are translates of each other. (d) $\mathbb{R}^2 \setminus \{(0,0)\}$; the circle of radius $e^{c/2}$ for every $c \in \mathbb{R}$.

1.2. (a) 0, since $0 \le \frac{x^2y^2}{x^2+y^2} \le y^2$. (b) Does not exist: the function is 1 on the x-axis and -1 on the y-axis. (c) 0: $|x|^3 \le |x|(x^2 + y^2)$ and similarly for y, so the absolute value is at most $|x| + |y|$. (d) 0: $|xy| \le \frac{1}{2}(x^2 + y^2)$, so the absolute value is at most $\frac{1}{2}\sqrt{x^2 + y^2}$.

1.3. For $\mathbf{v} = (a, b)$ with $a \neq 0$, $f(ta, tb) = \frac{tab^2}{a^2+t^2b^4} \to 0$; for $a = 0$, $f = 0$. On the parabola $x = y^2$, $f(y^2, y) = \frac{y^4}{2y^4} = \frac{1}{2}$ for $y \neq 0$.

1.4. (a) With $\sum x_i = 6$, $\sum x_i^2 = 14$, $\sum y_i = 10$, $\sum x_iy_i = 23$, $\sum y_i^2 = 38$: $L = \frac{1}{3}(14w^2 + 12wb + 3b^2 - 46w - 20b + 38)$. (b) $L(\mathbf{p}) = \mathbf{p}^\top A\mathbf{p} - 2\mathbf{c}^\top\mathbf{p} + d$ with $A = \frac{1}{3}\begin{pmatrix} 14 & 6 \\ 6 & 3 \end{pmatrix}$, which is positive definite ($14 > 0$, $\det = \frac{6}{9} > 0$). With $\mathbf{p}^* = A^{-1}\mathbf{c}$, $L(\mathbf{p}) = L(\mathbf{p}^*) + (\mathbf{p} - \mathbf{p}^*)^\top A(\mathbf{p} - \mathbf{p}^*)$, whose level sets are ellipses centered at $\mathbf{p}^*$, with axes along the eigenvectors of A.

Derivatives of Functions of One Variable

2.1. (a) Product rule and $\sigma' = \sigma(1 - \sigma)$. (b) $(\log\sigma)' = \frac{\sigma'}{\sigma} = 1 - \sigma$, and $1 - \sigma(x) = \frac{e^{-x}}{1+e^{-x}} = \frac{1}{e^x+1} = \sigma(-x)$. (c) Both one-sided difference quotients tend to 1 (the left one is $\frac{e^h-1}{h} \to 1$), so $f'(0) = 1$; $f'(x) = 1$ for $x \ge 0$ and $f'(x) = e^x$ for $x < 0$. The derivative f' is continuous, but not differentiable at 0: its one-sided derivatives are $e^0 = 1$ (left) and 0 (right). (d) The right quotient is 1, the left one 0.1; not differentiable at 0.

2.2. (a) $T_3(x) = 1 + \frac{x}{2} - \frac{x^2}{8} + \frac{x^3}{16}$. (b) $T_3(0.1) = 1.0488125$, while $\sqrt{1.1} = 1.0488088\ldots$. (c) $f^{(4)}(x) = -\frac{15}{16}(1 + x)^{-7/2}$, which has absolute value at most $\frac{15}{16}$ for $x \ge 0$, so the error is at most $\frac{15/16}{4!}(0.1)^4 \approx 3.9 \cdot 10^{-6}$ (the actual error is $3.65 \cdot 10^{-6}$).

2.3. (a) $\sin x - x = -\frac{x^3}{6} + (\text{terms of order } x^5)$, so the limit is $-\frac{1}{6}$. (b) $\frac{1}{2}$. (c) $\frac{1}{2}$. (d) $\log(1+x) - x = -\frac{x^2}{2} + \frac{x^3}{3} - \cdots$, so the limit is $-\frac{1}{2}$.

2.4. $f(3) - f(0) = 3f'(\xi) \ge -3$ for some $\xi \in (0, 3)$, so $f(0) \le f(3) + 3 = 8$. The function $f(x) = 8 - x$ attains this.

2.5. (a) Forward: 1.051709, error 0.0517, bound $\frac{e^{0.1}}{2}(0.1) \approx 0.0553$. Central: 1.001668, error 0.00167, bound $\frac{e^{0.1}}{6}(0.1)^2 \approx 0.00184$. (b) The errors are 0.0254 and 0.000417: they shrink by the factors 2.03 and 4.00, since the errors are approximately $\frac{f''(0)}{2}h$ and $\frac{f'''(0)}{6}h^2$.

Partial Derivatives and Gradients

3.1. (a) $(2xy^3-4y,\, 3x^2y^2-4x)^\top = (2,-1)^\top$. (b) $\frac{2\mathbf{x}}{\|\mathbf{x}\|^2} = \frac{1}{9}(2,4,4)^\top$. (c) $2(\mathbf{a}^\top\mathbf{x})\mathbf{a}$. (d) $-e^{-\|\mathbf{x}\|^2/2}\mathbf{x}$. (e) $2A\mathbf{x}+\mathbf{b}$.

3.2. (a) $(6xy,\, 3x^2-3y^2)^\top = (12,-9)^\top$. (b) The direction is $\frac{1}{5}(3,4)^\top$, and the directional derivative is $\frac{1}{5}(36-36)=0$. (c) Fastest increase in the direction $\frac{1}{5}(4,-3)^\top$ with rate $\|\nabla f(P)\| = 15$; fastest decrease in the direction $\frac{1}{5}(-4,3)^\top$ with rate -15. (d) $\pm\frac{1}{5}(3,4)^\top$, the directions tangent to the level curve through P.

3.3. (a) $\nabla T(2,1) = (-2x,-4y)^\top = (-4,-4)^\top$: move in the direction $-\frac{1}{\sqrt{2}}(1,1)^\top$, towards the origin; the temperature increases at the rate $4\sqrt{2} \approx 5.66$ per unit length. (b) The tangent vectors of $y=\frac{x^2}{4}$ are multiples of $(1,\frac{x}{2})$, and $\nabla T(x,\frac{x^2}{4}) = (-2x,-x^2)^\top = -2x(1,\frac{x}{2})^\top$. The bug ends up at the hottest point, the origin, where $T=100$.

3.4. (a) $\frac{x^2}{x^2+y^2} \le 1$, so $|f(x,y)| \le |y| \to 0$. (b) $\frac{f(ta,tb)}{t} = \frac{a^2b}{a^2+b^2} = a^2b$. (c) Both partial derivatives are 0 (take $\mathbf{v}=\mathbf{e}_1,\mathbf{e}_2$). If f were differentiable, all directional derivatives would be $\nabla f(0,0)^\top\mathbf{v}=0$, but $D_{\mathbf{v}}f(0,0) = \frac{1}{2\sqrt{2}}$ for $\mathbf{v} = \frac{1}{\sqrt{2}}(1,1)^\top$.

3.5. (a) $Df(1,1)\mathbf{h} = 2h_1+2h_2$ and $\langle\mathbf{g},\mathbf{h}\rangle_A = g_1h_1+4g_2h_2$, so $\mathbf{g} = (2,\frac{1}{2})^\top = A^{-1}\nabla f(1,1)$. (b) By the Cauchy–Schwarz inequality for $\langle\cdot,\cdot\rangle_A$, $\langle\mathbf{g},\mathbf{h}\rangle_A \le \|\mathbf{g}\|_A\|\mathbf{h}\|_A = \sqrt{5}$, with equality only for $\mathbf{h} = \mathbf{g}/\|\mathbf{g}\|_A$. (c) $\nabla f(1,1) = (2,2)^\top$ is the steepest direction for the Euclidean norm. In the norm $\|\cdot\|_A$, steps in the x_2-direction are twice as long, so the steepest direction leans towards x_1.

Jacobians: Derivatives of Vector-Valued Functions

4.1. (a) $DF = \begin{pmatrix} 2x & -2y \\ 2y & 2x \end{pmatrix}$, $\det DF = 4(x^2+y^2)$. (b) $F(1,2) = (-3,4)^\top$ and $DF(1,2) = \begin{pmatrix} 2 & -4 \\ 4 & 2 \end{pmatrix}$, so $F(1.1,1.9) \approx (-3,4)^\top + (0.6,0.2)^\top = (-2.4,4.2)^\top$; exact: $(-2.4,4.18)^\top$. (c) By the factor $4(x^2+y^2) = 4|z|^2$; areas are preserved locally on the circle $x^2+y^2=\frac{1}{4}$. (The Jacobian is $2|z|$ times a rotation, so F maps small squares to squares.)

4.2. (a) The diagonal matrix with entries 1 where $x_i > 0$ and 0 where $x_i < 0$. (b) $Df(\mathbf{x}) = \frac{1}{\|\mathbf{x}\|}I - \frac{\mathbf{x}\mathbf{x}^\top}{\|\mathbf{x}\|^3} = \frac{1}{\|\mathbf{x}\|}(I-\mathbf{u}\mathbf{u}^\top)$, and $(I-\mathbf{u}\mathbf{u}^\top)\mathbf{x} = \mathbf{x}-\mathbf{u}(\mathbf{u}^\top\mathbf{x}) = \mathbf{0}$. Expected, since $f(t\mathbf{x}) = f(\mathbf{x})$ for $t>0$: f is constant along the ray through $\mathbf{x}$. (c) $\frac{1}{T}\big(\operatorname{diag}(\mathbf{s}) - \mathbf{s}\mathbf{s}^\top\big)$ with $\mathbf{s} = \operatorname{softmax}(\mathbf{z}/T)$, by the chain rule.

4.3. (a) $\int_0^{2\pi}\int_0^1 r^2\cdot r\,dr\,d\varphi = \frac{\pi}{2}$. (b) $2\pi\big[-e^{-r^2/2}\big]_0^\infty = 2\pi$. (c) $\frac{1}{2\pi}\int_0^{2\pi}\int_0^r e^{-s^2/2}s\,ds\,d\varphi = 1-e^{-r^2/2}$, which is 0.95 for $r=\sqrt{2\log 20} \approx 2.45$. In two dimensions, more of the probability lies far from the center than in one dimension.

The Chain Rule

5.1. $Df(u,v) = (2u, -2)$ and $Dg(x,y) = \begin{pmatrix} 1 & 1 \\ y & x \end{pmatrix}$, so $D(f \circ g) = (2(x+y) - 2y,\ 2(x+y) - 2x) = (2x, 2y)$. Directly: $f(g(x,y)) = (x+y)^2 - 2xy = x^2 + y^2$.

5.2. (a) With $z = y\,\mathbf{w}^\top \mathbf{x}$: $\frac{d\ell}{dz} = \frac{-e^{-z}}{1+e^{-z}} = -\sigma(-z)$ and $Dz(\mathbf{w}) = y\,\mathbf{x}^\top$ $(1 \times d)$, so $\nabla \ell(\mathbf{w}) = -y\,\sigma(-y\,\mathbf{w}^\top \mathbf{x})\,\mathbf{x}$. (b) $X^\top(X\mathbf{w} - \mathbf{y}) + \lambda \mathbf{w}$, from the row vector $(X\mathbf{w} - \mathbf{y})^\top X$ of size $(1 \times N)(N \times d)$. (c) $\frac{A^\top(A\mathbf{x} - \mathbf{b})}{\|A\mathbf{x} - \mathbf{b}\|}$, from $\nabla \|\mathbf{r}\| = \frac{\mathbf{r}}{\|\mathbf{r}\|}$ and $D(A\mathbf{x} - \mathbf{b}) = A$.

5.3. (a) $\nabla T = (2xy, x^2)^\top$ and $\mathbf{x}'(t) = (-\sin t, \cos t)^\top$, so $\frac{d}{dt} T = -2\cos t \sin^2 t + \cos^3 t$, which is $\cos t(\cos^2 t - 2\sin^2 t)$ and the derivative of $\cos^2 t \sin t$. At $t = \frac{\pi}{4}$: $-\frac{\sqrt{2}}{4} \approx -0.354$. (b) $\cos t = 0$ (where $T = 0$), or $\tan^2 t = \frac{1}{2}$. The maximum $\frac{2}{3\sqrt{3}} \approx 0.385$ is attained where $\cos^2 t = \frac{2}{3}$ and $\sin t = \frac{1}{\sqrt{3}}$, i.e., at $(\pm\sqrt{2/3}, \frac{1}{\sqrt{3}})$.

5.4. (a) $\nabla = (2x, 8y) = (4, 8)$: $4(x-2) + 8(y-1) = 0$, i.e., $x + 2y = 4$. (b) $x + 2y + 2z = 9$. (c) $\nabla F = (2, 4, -1)$: $2(x-1) + 4(y-2) - (z-5) = 0$, i.e., $z = 2x + 4y - 5$.

5.5. $2x + y + xy' + 3y^2 y' = 0$; at $(1,1)$: $3 + 4y' = 0$, so $y'(1) = -\frac{3}{4}$. In general, $\frac{d}{dx} F(x, y(x)) = \frac{\partial F}{\partial x} + \frac{\partial F}{\partial y} y'(x) = 0$ by the chain rule.

Gradients with Respect to Matrices

6.1. (a) $\operatorname{tr}(A(X+H)^\top) = \operatorname{tr}(AX^\top) + \operatorname{tr}(AH^\top)$, and $\operatorname{tr}(AH^\top) = \sum_{i,j} A_{ij} H_{ij} = \langle A, H \rangle$. (b) $\|(X+H)\mathbf{a}\|^2 = \|X\mathbf{a}\|^2 + 2(X\mathbf{a})^\top H\mathbf{a} + \|H\mathbf{a}\|^2$, and $(X\mathbf{a})^\top H\mathbf{a} = \langle X\mathbf{a}\mathbf{a}^\top, H \rangle$. (c) $\operatorname{tr}((X+H)^2) = \operatorname{tr}(X^2) + 2\operatorname{tr}(XH) + \operatorname{tr}(H^2)$, and $\operatorname{tr}(XH) = \langle X^\top, H \rangle$. (d) Chain rule: $\|X\|_F = (\|X\|_F^2)^{1/2}$, so the gradient is $\frac{1}{2\|X\|_F} \cdot 2X$.

6.2. (a) The linear layer example of the lecture with $\boldsymbol{\delta} = W\mathbf{x} - \mathbf{y}$, plus $\nabla \frac{\lambda}{2} \|W\|_F^2 = \lambda W$. (b) $W^*\mathbf{x} = \frac{\|\mathbf{x}\|^2}{\|\mathbf{x}\|^2 + \lambda} \mathbf{y}$, so $(W^*\mathbf{x} - \mathbf{y})\mathbf{x}^\top + \lambda W^* = (\frac{\|\mathbf{x}\|^2}{\|\mathbf{x}\|^2 + \lambda} - 1 + \frac{\lambda}{\|\mathbf{x}\|^2 + \lambda}) \mathbf{y}\mathbf{x}^\top = 0$. (c) $W^*\mathbf{x} \to \mathbf{y}$ as $\lambda \to 0$ (a perfect fit), and $W^* \to 0$ as $\lambda \to \infty$: the regularization shrinks the weights.

6.3. (a) $\nabla \det(X) = \begin{pmatrix} d & -c \\ -b & a \end{pmatrix}$, and $X^{-1} = \frac{1}{\det X} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$, so $\det(X) X^{-\top} = \begin{pmatrix} d & -c \\ -b & a \end{pmatrix}$. (b) $\det(X) > 0$, so $\log\det X = \log|\det X|$ has the gradient $X^{-\top} = X^{-1}$, since X^{-1} is symmetric.

6.4. (a) $\mathbf{p} \approx (0.090, 0.245, 0.665)^\top$, $\ell \approx 0.408$, $\nabla \ell \approx (0.090, 0.245, -0.335)^\top$. (b) Chain rule with $D(\mathbf{z}/T) = \frac{1}{T} I$. For $T = \frac{1}{2}$: $\mathbf{p} \approx (0.016, 0.117, 0.867)^\top$, $\ell \approx 0.143$, $\nabla \ell \approx (0.032, 0.235, -0.266)^\top$. For $T = 2$: $\mathbf{p} \approx (0.186, 0.307, 0.506)^\top$, $\ell \approx 0.680$, $\nabla \ell \approx (0.093, 0.154, -0.247)^\top$.

Backpropagation and Automatic Differentiation

7.1. Values: $v_1 = 2$, $v_2 = 0$, $v_3 = 1$, $f = v_4 = 2$. Forward mode with $(\dot{x}, \dot{y}) = (1, 0)$: $\dot{v}_1 = 1$, $\dot{v}_2 = \dot{x}y + x\dot{y} = 0$, $\dot{v}_3 = e^{v_2}\dot{v}_2 = 0$, $\dot{v}_4 = \dot{v}_1 v_3 + v_1 \dot{v}_3 = 1$; with $(0, 1)$: $\dot{v}_1 = 1$, $\dot{v}_2 = 2$, $\dot{v}_3 = 2$, $\dot{v}_4 = 1 + 4 = 5$. Reverse mode: $\bar{v}_4 = 1$, $\bar{v}_1 = v_3 = 1$, $\bar{v}_3 = v_1 = 2$, $\bar{v}_2 = \bar{v}_3 e^{v_2} = 2$, $\bar{x} = \bar{v}_1 + \bar{v}_2 y = 1$, $\bar{y} = \bar{v}_1 + \bar{v}_2 x = 5$. So $\nabla f(2, 0) = (1, 5)^\top$; directly, $\frac{\partial f}{\partial x} = e^{xy}(1 + (x+y)y)$ and $\frac{\partial f}{\partial y} = e^{xy}(1 + (x+y)x)$.

7.2. (a) $\mathbf{z} = (2, -0.5)^\top$, $\mathbf{a} = (2, 0)^\top$, $\hat{y} = 2.5$, $L = 0.5$. (b) $\hat{y} - y = 1$; $\nabla_{\mathbf{w}_2} L = \mathbf{a} = (2, 0)^\top$; $\frac{\partial L}{\partial b_2} = 1$; $\nabla_{\mathbf{a}} L = \mathbf{w}_2 = (1, -1)^\top$; $\nabla_{\mathbf{z}} L = (1, 0)^\top \odot (1, -1)^\top = (1, 0)^\top$; $\nabla_{W_1} L = \begin{pmatrix} 1 & 0.5 \\ 0 & 0 \end{pmatrix}$; $\nabla_{\mathbf{b}_1} L = (1, 0)^\top$. (c) The second hidden unit is inactive ($z_2 < 0$): its incoming weights (the second row of W_1 and the second entry of $\mathbf{b}_1$) get zero gradients since $\mathrm{ReLU}'(z_2) = 0$, and its outgoing weight $(\mathbf{w}_2)_2$ gets a zero gradient since $a_2 = 0$. With only this example, these weights never change: a "dead" ReLU unit.

7.3. (a) $f : \mathbb{R}^{1000} \to \mathbb{R}$: 1000 forward passes or 1 reverse pass. $f : \mathbb{R} \to \mathbb{R}^{1000}$: 1 forward pass or 1000 reverse passes. (b) $J_2 J_1$ costs $100 \cdot 100 \cdot 10^4 = 10^8$, and then $J_3(J_2 J_1)$ another 10^6. $J_3 J_2$ costs 10^4, and then $(J_3 J_2) J_1$ another 10^6. So the second order, which is reverse mode, is about 100 times cheaper.

7.4. (a) By the chain rule, $f_L'(x)$ is a product of L values of σ', each at most $\frac{1}{4}$. For $L = 10$: $4^{-10} \approx 9.5 \cdot 10^{-7}$. (b) $g_{l+1}'(x) = \sigma'(w\, g_l(x))\, w\, g_l'(x)$, so each layer contributes a factor of absolute value at most $\frac{|w|}{4}$. Gradients can only avoid vanishing if $|w| > 4$, but large weights push σ into its flat regions, where σ' is tiny; this is one reason why deep networks use ReLU.

Higher Derivatives and Taylor's Theorem

8.1. (a) $\nabla f = (3x^2 - 2y^2,\, -4xy + 4y^3)^\top$ and $\nabla^2 f = \begin{pmatrix} 6x & -4y \\ -4y & -4x + 12y^2 \end{pmatrix}$; both mixed partial derivatives are $-4y$. (b) $f(1,1) = 0$, $\nabla f(1,1) = (1, 0)^\top$, $\nabla^2 f(1,1) = \begin{pmatrix} 6 & -4 \\ -4 & 8 \end{pmatrix}$, so $f(1 + h_1, 1 + h_2) \approx h_1 + 3h_1^2 - 4h_1 h_2 + 4h_2^2$. At $(0.1, -0.1)$: 0.21; exact: 0.2051. (c) $\frac{1}{2}(6 - 8 + 8) = 3$. The largest and smallest curvatures are the eigenvalues $7 \pm \sqrt{17}$, i.e., 11.12 and 2.88.

8.2. (a) $X^\top X + \lambda I$. (b) $\frac{1}{\|\mathbf{x}\|}(I - \mathbf{u}\mathbf{u}^\top)$ with $\mathbf{u} = \frac{\mathbf{x}}{\|\mathbf{x}\|}$ (the Jacobian of the gradient $\mathbf{u}$, Problem 4.2). Eigenvalue 0 with eigenvector $\mathbf{x}$, and eigenvalue $\frac{1}{\|\mathbf{x}\|}$ on the orthogonal complement of $\mathbf{x}$. Along the ray through $\mathbf{x}$, $\|t\mathbf{x}\| = t\|\mathbf{x}\|$ is linear, so there is no curvature in this direction. (c) $\sigma(\mathbf{w}^\top \mathbf{x})(1 - \sigma(\mathbf{w}^\top \mathbf{x}))\mathbf{x}\mathbf{x}^\top$, of rank 1 (for $\mathbf{x} \neq \mathbf{0}$).

8.3. (a) $\nabla f(0,0) = (\frac{1}{2}, \frac{1}{2})^\top$ and $\nabla^2 f(0,0) = \frac{1}{4}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$. (b) $\log 2 + \frac{1}{2}(0.2 - 0.1) + \frac{1}{8}(0.2 + 0.1)^2 \approx 0.754397$; exact: 0.754355. (c) Eigenvalue $\frac{1}{2}$ with eigenvector $(1, -1)$, and 0 with eigenvector $(1, 1)$: along $(1, 1)$, $f(t, t) = \log 2 + t$ is linear.

8.4. (a) $f(x, y) = x\, g(x, y)$ with $g = \frac{y(x^2 - y^2)}{x^2 + y^2}$, and $f(0, y) = 0$, so $\frac{\partial f}{\partial x}(0, y) = \lim_{h \to 0} g(h, y) = -y$ (also for $y = 0$). Similarly $f = y\, k$ with $k = \frac{x(x^2 - y^2)}{x^2 + y^2}$ gives $\frac{\partial f}{\partial y}(x, 0) = x$. (b) Differentiate $-y$ with respect to y, and x with respect to x. Schwarz's theorem needs continuous second partial derivatives, and they are not continuous at the origin.

Unconstrained Optimization

9.1. (a) $\nabla f = (2x + y - 3,\, x + 2y)^\top = \mathbf{0}$ at $(2, -1)$. The Hessian $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ has the eigenvalues 1 and 3: a strict local minimum with $f = -3$, which is global (a quadratic function with a positive definite

Hessian). (b) $\nabla f = (3x^2 - 3,\, 2y - 2)^\top$, so the critical points are $(\pm 1, 1)$, and $\nabla^2 f = \mathrm{diag}(6x, 2)$. At $(1,1)$: eigenvalues 6 and 2, a strict local minimum with $f = -3$, not global since $f(x,1) \to -\infty$ as $x \to -\infty$. At $(-1,1)$: eigenvalues -6 and 2, a saddle point with $f = 1$.

9.2. (a) $\nabla f = \big(y(2x + y - 3),\, x(x + 2y - 3)\big)^\top$ vanishes at $(0,0)$, $(3,0)$, $(0,3)$ and $(1,1)$. With $\nabla^2 f = \begin{pmatrix} 2y & 2x + 2y - 3 \\ 2x + 2y - 3 & 2x \end{pmatrix}$: at the first three points, $\det = -9 < 0$ (saddle points); at $(1,1)$, the eigenvalues are 1 and 3 (a strict local minimum with $f = -1$). (b) The lines $x = 0$, $y = 0$ and $x + y = 3$ bound a triangle on which $f < 0$, and f changes sign across each line. Where two of the lines cross, f has alternating signs in the four sectors, as xy does at the origin, which gives a saddle point. In the triangle, f is negative and 0 on the boundary, so it has a minimum inside.

9.3. (a) $\nabla^2 f = \begin{pmatrix} 6 & 2 \\ 2 & 6 \end{pmatrix}$: eigenvalue 8 for $(1,1)$ and 4 for $(1,-1)$; indeed $f(t,t) = 8t^2$ and $f(t,-t) = 4t^2$. The level curves are ellipses, longest in the direction $(1,-1)$ of the smaller curvature, with axis ratio $\sqrt{8/4} = \sqrt{2}$. (b) $\nabla^2 g = \begin{pmatrix} 2 & 6 \\ 6 & 2 \end{pmatrix}$: eigenvalue 8 for $(1,1)$ and -4 for $(1,-1)$, so the origin is a saddle point; $g(t,t) = 8t^2$ increases and $g(t,-t) = -4t^2$ decreases. $g = 0$ on the lines $y = (-3 \pm 2\sqrt{2})x$, i.e., $y \approx -0.17x$ and $y \approx -5.83x$; the other level curves are hyperbolas with these lines as asymptotes.

9.4. (a) $\nabla L = \frac{1}{3}(28w + 12b - 46,\, 12w + 6b - 20)^\top$; $\nabla L = \mathbf{0}$ gives $14w + 6b = 23$ and $6w + 3b = 10$, so $(w, b) = (\frac{3}{2}, \frac{1}{3})$. (b) $\nabla^2 L = \frac{2}{3}\begin{pmatrix} 14 & 6 \\ 6 & 3 \end{pmatrix}$, with eigenvalues $\frac{17 \pm \sqrt{265}}{3} \approx 11.09$ and 0.240. Both are positive, and L is a quadratic function with this constant Hessian, so the critical point is the unique global minimum (the example on quadratic functions in the lecture). (c) The residuals are $-\frac{1}{6}, \frac{1}{3}, -\frac{1}{6}$, so $L^* = \frac{1}{18} \approx 0.056$. The normal equations $\Phi^\top \Phi \boldsymbol{\theta} = \Phi^\top \mathbf{y}$ with the rows $(x_i, 1)$ of Φ are the same two equations.

9.5. (a) $f = y^2 - 3x^2 y + 2x^4$, $\nabla f = (-6xy + 8x^3,\, 2y - 3x^2)^\top$ vanishes at the origin, and $\nabla^2 f(0,0) = \mathrm{diag}(0, 2)$ has the eigenvalue 0. (b) On $y = mx$: $f = x^2(m - x)(m - 2x) \approx m^2 x^2$ for $m \neq 0$; on the x-axis $f = 2x^4$, and on the y-axis $f = y^2$. (c) $f(x, \frac{3}{2}x^2) = \frac{x^2}{2} \cdot (-\frac{x^2}{2}) = -\frac{x^4}{4} < 0$ for $x \neq 0$: between the two parabolas $y = x^2$ and $y = 2x^2$, f is negative.

Gradient Descent and Newton's Method

10.1. (a) $\nabla f = (4x, 16y)^\top$: $(1,1) \to (0.6, -0.6) \to (0.36, 0.36)$. (b) The Hessian has the eigenvalues 4 and 16, so gradient descent converges if and only if $0 < \eta < \frac{1}{8}$; the best step size is $\eta = \frac{2}{4+16} = 0.1$, with $\rho = \frac{\kappa - 1}{\kappa + 1} = 0.6$ for $\kappa = 4$. (c) $0.6^k \leq 10^{-6}$ for $k \geq 28$. (d) y is multiplied by $1 - 16 \cdot 0.125 = -1$ in every step, so it jumps between $\pm y_0$ forever, while x converges.

10.2. (a) With the eigenvalues $\lambda_1 \approx 11.09$ and $\lambda_2 \approx 0.240$: convergence for $0 < \eta < \frac{2}{\lambda_1} \approx 0.180$; the best step size is $\frac{2}{\lambda_1 + \lambda_2} = \frac{3}{17} \approx 0.176$, with $\kappa \approx 46.1$ and $\rho \approx 0.958$, so about 319 steps. (b) The new Hessian is $\frac{2}{3}\begin{pmatrix} \sum (x_i - 2)^2 & \sum (x_i - 2) \\ \sum (x_i - 2) & 3 \end{pmatrix} = \mathrm{diag}(\frac{4}{3}, 2)$, with $\kappa = 1.5$; the best step size is 0.6, with $\rho = 0.2$, so 9 steps. Since $w(x - 2) + c = wx + (c - 2w)$, we get $b = c - 2w$ (indeed $c^* = \frac{10}{3}$, $w^* = \frac{3}{2}$, $b^* = \frac{1}{3}$). (c) After centering, the two columns $(x_i - 2)$ and $(1, 1, 1)$ of the design matrix

are orthogonal, so the Hessian is diagonal: the slope and the intercept no longer interact, and the level sets are almost circles.

10.3. (a) $f'(x) = 1 - \frac{1}{x}$ vanishes at $x = 1$, and $f''(x) = \frac{1}{x^2} > 0$, so f is convex and $f(1) = 1$ is the global minimum. (b) $x - \frac{f'(x)}{f''(x)} = x - (1 - \frac{1}{x})x^2 = 2x - x^2$, and $1 - (2x - x^2) = (1 - x)^2$. From 0.5: 0.75, 0.9375, 0.99609375, 0.99998474, with errors 0.25, 0.0625, 0.0039, $1.5 \cdot 10^{-5}$. (c) $1 - x_k = (1 - x_0)^{2^k}$, which tends to 0 if and only if $|1 - x_0| < 1$, i.e., $0 < x_0 < 2$. For $x_0 = 3$, $x_1 = -3$ is not even in the domain.

10.4. $f'(x) = x^2 - 2$ and $f''(x) = 2x$, so $x - \frac{x^2-2}{2x} = \frac{1}{2}(x + \frac{2}{x})$. The iterates 1.5, 1.416667, 1.414216 converge to $\sqrt{2}$, with errors 0.086, 0.0025, $2.1 \cdot 10^{-6}$. For $\sqrt{a}$, minimize $\frac{x^3}{3} - ax$, which gives $x_{k+1} = \frac{1}{2}(x_k + \frac{a}{x_k})$.

10.5. (a) $s' = \sigma$ and $s'' = \sigma(1 - \sigma) \in (0, \frac{1}{4}]$. (b) With $\eta = \frac{1}{L} = 4$: $s(x_{k+1}) \le s(x_k) - 2\sigma(x_k)^2$, and $\min_{k<K} \sigma(x_k)^2 \le \frac{s(x_0)}{2K}$, since $s > 0$. The iterates $x_{k+1} = x_k - 4\sigma(x_k)$ decrease to $-\infty$, $s(x_k) \to 0 = \inf s$, and the gradients $\sigma(x_k) \to 0$. The theorem only promises small gradients, not a minimum, so this is consistent.

Convexity

11.1. (a) Convex, even strictly: the Hessian has the eigenvalues 1 and 3. (b) Not convex: $\begin{pmatrix} 2 & 4 \\ 4 & 2 \end{pmatrix}$ has the eigenvalue -2. (c) Convex: the Hessian $e^{\|\mathbf{x}\|^2}(2I + 4\mathbf{x}\mathbf{x}^\top)$ is positive definite. (d) Convex, like every norm. (e) Convex: $f''(x) = \frac{1}{4}x^{-3/2} > 0$. (f) Convex: the Hessian is $\frac{2}{y^3}\begin{pmatrix} y^2 & -xy \\ -xy & x^2 \end{pmatrix} = \frac{2}{y^3}\mathbf{v}\mathbf{v}^\top$ with $\mathbf{v} = (y, -x)^\top$, which is positive semidefinite. (g) Not convex: the Hessian $\begin{pmatrix} 0 & -1/y^2 \\ -1/y^2 & 2x/y^3 \end{pmatrix}$ has determinant $-\frac{1}{y^4} < 0$.

11.2. (a) $(\log x)'' = -\frac{1}{x^2} < 0$. (b) Concavity gives $\log(\theta a + (1 - \theta)b) \ge \theta \log a + (1 - \theta) \log b$; apply exp, which is increasing. (c) With $a = x$, $b = \frac{4}{x}$ and $\theta = \frac{1}{2}$: $x + \frac{4}{x} \ge 2\sqrt{4} = 4$, with equality for $x = \frac{4}{x}$, i.e., $x = 2$. Alternatively, the derivative $1 - \frac{4}{x^2}$ vanishes at $x = 2$, and the second derivative $\frac{8}{x^3}$ is positive, so the function is convex on $(0, \infty)$ and 2 is the global minimum.

11.3. (a) Differentiating $\nabla \ell = -y\,\sigma(-y\,\mathbf{w}^\top\mathbf{x})\,\mathbf{x}$ gives $\nabla^2 \ell = \sigma(1 - \sigma)\,y^2\,\mathbf{x}\mathbf{x}^\top = \sigma(1 - \sigma)\,\mathbf{x}\mathbf{x}^\top$ (with σ evaluated at $-y\,\mathbf{w}^\top\mathbf{x}$), which is positive semidefinite. (b) It is the maximum of the affine functions 0 and $1 - y\,\mathbf{w}^\top\mathbf{x}$. (c) $\|X\mathbf{w} - \mathbf{y}\|^2$ has the positive semidefinite Hessian $2X^\top X$, $\lambda\|\mathbf{w}\|_1$ is a nonnegative multiple of a norm, and sums of convex functions are convex. (d) $f(0) = 0$ and $f(2) = 1$, but $f(1) = 1 > \frac{1}{2}f(0) + \frac{1}{2}f(2)$.

11.4. $f(\mathbf{y}) \ge f(\mathbf{x}^*) + \nabla f(\mathbf{x}^*)^\top(\mathbf{y} - \mathbf{x}^*) \ge f(\mathbf{x}^*)$ for all $\mathbf{y} \in C$, by the first-order characterization of convexity. Picture: at a boundary point $\mathbf{x}^*$ of a half-plane, the steepest descent direction $-\nabla f(\mathbf{x}^*)$ points straight out of C, so we cannot decrease f without leaving C. (This is the KKT condition for one linear constraint.)

11.5. (a) $f \ge 0$, and $f = 0$ exactly on the hyperbola $w_1 w_2 = 1$. (b) $(1, 1)$ and $(-1, -1)$ are minima with $f = 0$, but their midpoint $(0, 0)$ has $f = 1$. (c) $\nabla f = 2(w_1 w_2 - 1)(w_2, w_1)^\top$ vanishes on the hyperbola and at the origin. At the origin, $\nabla^2 f = \begin{pmatrix} 0 & -2 \\ -2 & 0 \end{pmatrix}$ has the eigenvalues ± 2: a saddle

point. At $(1,1)$, $\nabla^2 f = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix}$ has the eigenvalues 4 (for $(1,1)$) and 0 (for $(1,-1)$, the tangent direction of the hyperbola, along which f stays 0).

Constrained Optimization

12.1. $(y,x)+\lambda(2x,8y)=(0,0)$ gives $y=-2\lambda x$ and $x=-8\lambda y=16\lambda^2 x$, so $\lambda=\pm\frac{1}{4}$ (the case $x=0$ forces $y=0$, which is not on the ellipse). For $\lambda=-\frac{1}{4}$: $y=\frac{x}{2}$, $x=\pm 2$, so the maximum is 2 at $\pm(2,1)$. For $\lambda=\frac{1}{4}$: the minimum is -2 at $\pm(2,-1)$. At $(2,1)$, $\nabla(xy)=(1,2)$ and $\nabla g=(4,8)$ are parallel, and both curves have the tangent line $x+2y=4$.

12.2. (a) $(yz,xz,xy)=\mu(1,1,1)$ gives $x=y=z=4$, and the maximum is 64. (b) The Lagrange conditions $y+2z=\mu yz$, $x+2z=\mu xz$, $2x+2y=\mu xy$ give $x=y$ and then $x=2z$, so $4z^3=32$: $x=y=4$, $z=2$, with surface area 48.

12.3. Optimize $\|\mathbf{x}\|^2$ subject to $\mathbf{x}^\top Q\mathbf{x}=3$ with $Q=\begin{pmatrix} 1 & 1/2 \\ 1/2 & 1 \end{pmatrix}$. The Lagrange condition $2\mathbf{x}-2\mu Q\mathbf{x}=\mathbf{0}$ says that $\mathbf{x}$ is an eigenvector of Q, and then $\|\mathbf{x}\|^2=\frac{3}{\text{eigenvalue}}$. Eigenvalue $\frac{3}{2}$ for $(1,1)$: the closest points $\pm(1,1)$, at distance $\sqrt{2}$. Eigenvalue $\frac{1}{2}$ for $(1,-1)$: the farthest points $\pm(\sqrt{3},-\sqrt{3})$, at distance $\sqrt{6}$.

12.4. $A=\begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & -1 \end{pmatrix}$, $AA^\top=\operatorname{diag}(3,2)$, $(AA^\top)^{-1}\mathbf{b}=(1,\frac{1}{2})^\top$, and $\mathbf{x}^*=A^\top(1,\frac{1}{2})^\top=(\frac{3}{2},1,\frac{1}{2})^\top$, with $\|\mathbf{x}^*\|^2=\frac{7}{2}$. The null space is spanned by $\mathbf{v}=(1,-2,1)^\top$, ${\mathbf{x}^*}^\top\mathbf{v}=0$, and $\|\mathbf{x}^*+t\mathbf{v}\|^2=\frac{7}{2}+6t^2$.

12.5. (a) Stationarity $2(x-3)+\mu=0$, $2(y-2)+\mu=0$. For $\mu=0$, the point $(3,2)$ is not feasible, so $x+y=2$, which gives $\mu=3\geq 0$ and $(x,y)=(1.5,0.5)$; this is the global minimum, since the problem is convex. It is the projection of $(3,2)$ onto the half-plane, at distance $\frac{3}{\sqrt{2}}=\frac{|3+2-2|}{\sqrt{2}}$. (b) The minimum is $(0,2)$. The constraints $x+y\leq 2$ and $-x\leq 0$ are active, with $\mu_1=\mu_2=6$, and $-y\leq 0$ is inactive, with $\mu_3=0$: indeed $\nabla f(0,2)=(0,-6)$ and $(0,-6)+6(1,1)+6(-1,0)=(0,0)$.

12.6. With the half-sides $x,y,z>0$, maximize $8xyz$ subject to $x^2+y^2+z^2=1$. The Lagrange conditions give $x=y=z=\frac{1}{\sqrt{3}}$: a cube with side $\frac{2}{\sqrt{3}}\approx 1.155$ and volume $\frac{8}{3\sqrt{3}}\approx 1.540$, which is the fraction $\frac{2}{\sqrt{3}\,\pi}\approx 36.8\%$ of the volume $\frac{4\pi}{3}$ of the ball.